

Modeling of the mixing of monodisperse particles using a stationary DEM-based Markov process

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Abstract

This paper provides a discussion on the modeling of granular mixing using Markov chain theory. Previous papers on this topic are either based on restrictive underlying assumptions about the flow structure or are limited to a small number of states. In this paper, a generalized approach for the construction of a multidimensional state space first-order Markov chain that represents the mixing of monodisperse particles is introduced. The transition probability matrix is computed directly using results obtained from a discrete element model. This work shows that, if accurate measurements of the state of the system are available, the associated Markov operator leads to a good estimate of the particle dynamics in the system.

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1. Introduction

Current trends in modeling of particle mixing are based on complex and computationally intensive discrete element methods (DEM) (Bertrand, Leclaire, & Levecque, 2005). These models are experiencing rapid development and an increasing body of experiments tends to demonstrate their ability to predict several macroscopic quantities such as the evolution of the index of segregation, the torque acting on the shell as well as the mean velocity profiles in blenders (Kuo et al., 2002; Lemieux et al., 2006; Sudah, Arratia, Alexander, & Muzzio, 2005). However, the large computational resources needed for the simulation of realistic industrial mixing applications using these models constitute a major limitation and motivate the development of other modeling approaches.

One such modeling approach is based on Markov processes, which have been rarely used for analysis and modeling of granular mixing. In fact, an early introduction of such Markov process based models by Chen and Fan (1972) for the simulation of gran-

ular mixing in static mixers showed interesting results based on accurate and reproducible experimental measurements of the state transitions. More recently, a similar model was used to capture the evolution of mixing in a hoop mixer (Aoun-Habbache, Aoun, Berthiaux, & Mizonov, 2002) in both radial and axial directions. However, due to the difficulty of constructing a robust Markov transition matrix, as a result of poor tracking methods, restrictive underlying assumptions were made about the motion of the particles by imposing the set of possible transitions as well as by constructing the transition probabilities based on two degrees of freedom. Nevertheless, the results obtained with these simplifications brought sufficient evidence that Markov process based models constitute an interesting avenue for modeling granular mixing, as reviewed in Berthiaux, Mizonov, and Zhukov (2005).

The objective of this paper is to introduce a new modeling strategy for the simulation of the mixing of monodisperse particles. The proposed method is based on the assumption that the flow and the mixing of monodisperse particles can be modeled by a stationary Markov process, that is by an underlying state transition matrix that is independent of time. This process is built by directly measuring the state transitions of the particles over a finite state space using DEM results. A similar assumption was made in the work of Chen and Fan (1972), Aoun-Habbache et

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al. (2002) and Chou, Johnson, and Rippie (1977). Note that a few reports from the literature have used nonstationary Markov chains for describing the flow of monodisperse and polydisperse systems of particles (Chou et al., 1977; Rippie & Chou, 1978). It should be mentioned that, up to now, most of the DEM simulations from the literature on granular mixing have been done for monodisperse systems (Bertrand et al., 2005). Despite this fact, this large body of work has helped gain insight into the mixing behavior and performance of various blenders. As will be seen, the proposed approach has the advantage of being much less computationally intensive than common discrete element methods so it has the potential to handle larger and more realistic systems of monodisperse particles.

The paper is divided as follows. First, a brief definition of Markov process is presented. Next, the procedure for the construction of a Markov chain using a time series of the location of particles, using a few iterations of a DEM simulation, is described in detail. Finally, the proposed approach is applied to simulate the mixing of spherical monosized particles in a drum tumbler. More precisely, the effect of several parameters of the method on the solution is addressed, namely the time step of the process, the number of states and the learning time, the latter being the time of DEM simulation used for building the transition matrix. In particular, this work shows that, under certain conditions, the mixing dynamics of systems with an large number of degrees of freedom can be predicted with good accuracy by a linear map within the framework of Markov processes.

2. Methodology

2.1. Definition of the Markov process

Given a probability space Π , a finite N -dimensional state space $S = \{1, 2, \dots, N\}$, a stochastic process $\{X_n\}_{n \in \mathbb{N}}$ and an invariant probability measure $\mathcal{P}$ on Π . The stochastic process is a Markov process if

$$\mathcal{P}\{X_{n+1}|X_n, X_{n-1}, \dots, X_0\} = \mathcal{P}\{X_{n+1}|X_n\}. \quad (1)$$

In other words, a Markovian evolution X_{n+1} depends only on the current state of the system X_n (Markov property) (Chung, 1967). By construction, the probability $\mathcal{P}(X_0 = j) = 1$ for some $j \in S$ when X starts from state j . More generally, the probability of transition from state j to state i , $p^{(n)}(i|j)$, after n iterations of the map, is defined as

$$p^{(n)}(i|j) = \mathcal{P}\{X_n = i | X_{n-1} = j\}, \quad (2)$$

for all $i, j \in S$. In this work, the process is assumed to be stationary, i.e. the transition probabilities are independent of n , so that $p^{(n)}(i|j) = p(i|j)$.

Let us consider the N -dimensional state vector after l iterations, $\mu_l = [\mu_l(1), \mu_l(2), \dots, \mu_l(N-1), \mu_l(N)]$, such that each element is given by

$$\mu_l(i) = \mathcal{P}\{X_l = i\}, \quad \forall i \in S, 0 \leq l \leq (n+1). \quad (3)$$

The Markov property implies that the state of the chain can be written as

$$\mu_{n+1}(i_{n+1}) = \mu_0(i_0) \prod_{k=1}^n p(i_k | i_{k-1}), \quad (4)$$

where $\{i_0, i_1, \dots, i_n, i_{n+1}\} \in S^{n+2}$ represents a sequence of events up to iteration $n+1$. By definition, the stochastic transition matrix $\mathbf{P}$, defined as $\mathbf{P}_{ij} = p(j|i)$, has the following properties:

$$\mathbf{P}_{ij} \geq 0, \quad \sum_{i \in S} \mathbf{P}_{ij} = 1, \quad \forall j \in S. \quad (5)$$

This equation along with Eq. (4) then leads to

$$\mu_n = \mu_0 \mathbf{P}^n. \quad (6)$$

2.2. Construction of the Markov process

It follows from Eq. (6) that the construction of the Markov process entails setting the value of each entry of the transition matrix $\mathbf{P}$. In this work, it is proposed to use the first few iterations of a DEM simulation to predict the position of the particles in the system and define this transition matrix. The reader is referred to a recent review by Bertrand et al. (2005) for more details on DEM and its use in granular mixing. The DEM model used for the purpose of this work is a variant of the spring-dashpot model of Cundall and Strack (1979) with tangential and normal components and is described elsewhere (Lemieux et al., in press). In this DEM model, the particles are spherical and free to rotate.

Let us consider a decomposition of the computational domain Ω into an unstructured mesh of N tetrahedra forming the N states of S . After each time interval τ in the DEM simulation, which corresponds to a given number of time steps, the current position of each particle originally in state i is considered in order to determine its current state. The total number of particles in state $i \in S$ at time t is first computed as

$$\varphi(i, t) = \sum_{p=1}^{N_p} \phi_p(i, t), \quad (7)$$

where N_p is the total number of particles in Ω and $\phi_p(i, t)$ is an indicator function that returns 1 if particle p is in state i at time t and 0 otherwise. The probability of a transition from state i to state j , $\mathbf{P}_{ij}$, can then be obtained by

$$\mathbf{P}_{ij} = \frac{1}{N_{LT}} \sum_{n=1}^{N_{LT}} \frac{1}{\varphi(i, t_n)} \sum_{p=1}^{N_p} \phi_p(j, t_n) \phi_p(i, t_{n-1}), \quad (8)$$

where $t_n = n\tau$ and $T = N_{LT}\tau$. In these expressions, the time interval τ in the DEM simulation is in fact the Markov process time step, T represents the learning time for the construction of the transition matrix, i.e. the time during which the DEM simulation is run for the evaluation of the transition probabilities, and N_{LT} is the number of time intervals τ required for the learning process. Note that the construction of the matrix $\mathbf{P}$ begins when permanent regime conditions prevail in the mixing system.

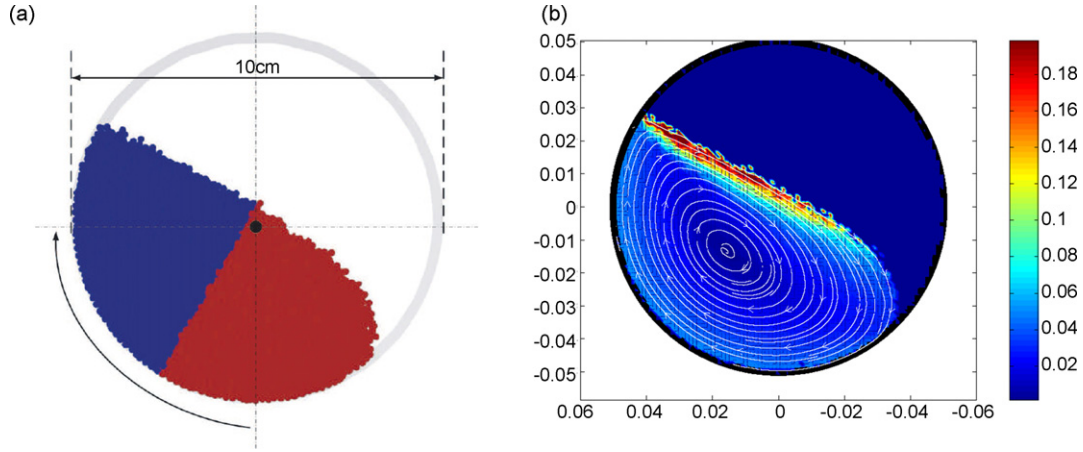


Fig. 1. (a) Initial particle distribution in the mixing system. (b) Mean velocity field and streamlines in permanent regime (legend showing the magnitude of the velocity in the radial plane in m/s).

In the remainder of the paper, the accuracy of the proposed DEM-based Markov process method (DEM-MPM) is assessed by means of a standard benchmark problem in granular mixing. In particular, the effect of the Markov process time step τ , the number of states N and the learning time T on the transition matrix is investigated.

2.3. Application to granular mixing

Let us consider a mixing system containing, for instance, red and blue particles with identical properties. A color function $C(p)$, $p = 1, \dots, N_p$, is introduced to indicate the color of particle p . In this case, with no loss of generality, this function takes two values, 1 or 0, corresponding to blue and red respectively. Starting with an initial color distribution such as the one illustrated in Fig. 1(a), the successive positions of the particles over time are computed using the Markov process described in the previous section.

For comparison purposes, the DEM solution is mapped onto S and the fraction of colored material in each state is computed. The relative standard deviation (RSD) of the color fraction distribution over S is then taken as a measure of mixing. This index is obtained by first computing the variance σ^2 of the fraction of one particle type (say red) in each state:

$$\sigma^2 = \frac{1}{N-1} \sum_{j \in S} (x_j - \bar{x})^2, \quad (9)$$

Table 1
Parameters used for the DEM-based Markov process method

Parameter (units)	Value
Number of particles	48,000
Particle diameter (mm)	2.0
Drum velocity (RPM)	10
Time step (DEM) (s)	1.5×10^{-5}
τ (s)	1/12, 1, 2, 3, 4
N_{LT} (iterations)	1–12
N (mesh size) (cells (cm))	72 (3.0), 253 (2.0), 1813 (1.0), 3587 (0.75), 5595 (0.60)

where x_j is the fraction of red particles in cell j and $\bar{x}$ is the total fraction of red particles in Ω . The RSD is then given by

$$\text{RSD} = \frac{\sigma}{\bar{x}}. \quad (10)$$

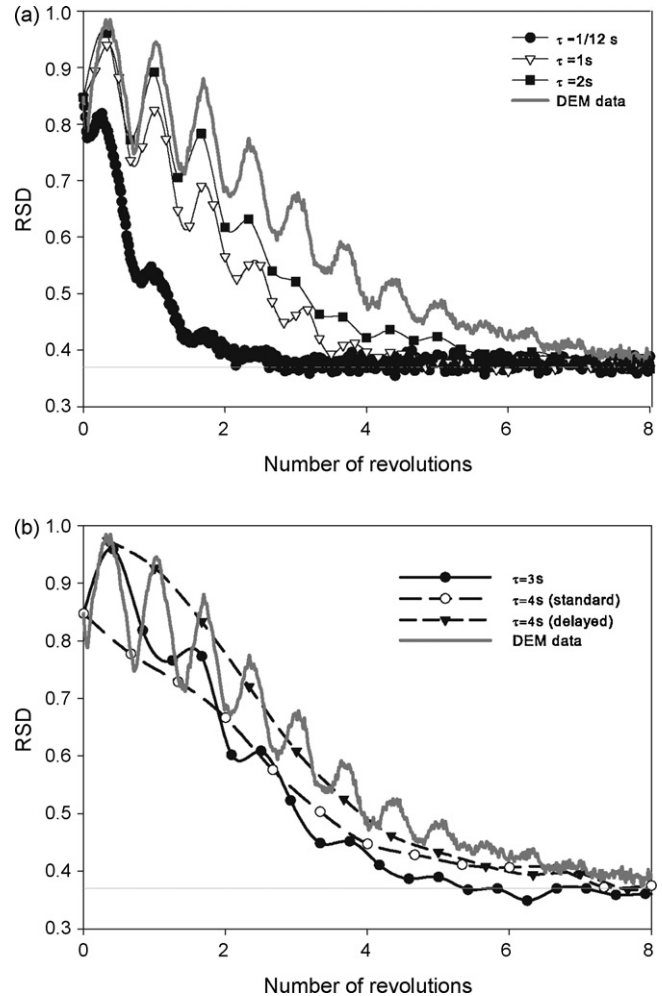


Fig. 2. Comparison of RSD curves obtained using DEM and those obtained with DEM-MPM with $N = 1813$, $N_{LT} = 2$ and different values of τ .

It is well known that a lower bound for the RSD is the standard deviation for a completely random binary mixture (Danckwerts, 1952):

$$\text{RSD}_r = \frac{1}{\bar{x}} \sqrt{\frac{\bar{x}(1-\bar{x})}{m}}, \quad (11)$$

where m is the number of particles in each cell. In the benchmark problem considered in this work (see Section 3), m is not fixed, therefore Eq. (11) was not used to evaluate RSD_r . Instead, it was approximated by calculating the RSD value of one hundred random binary mixtures using Eqs. (9) and (10), which yielded an RSD_r of 0.37 ± 0.02 .

Finally, one comment must be made concerning the implementation of the Markov process introduced in this work. If N_i represents the number of particles initially in state i , then the application of one step of the Markov process will involve fractions of particles of a given type (or color) that are multiples of $1/N_i$. Therefore, to avoid dealing with non-integer numbers of particles when using the Markov process, which would render impossible the comparison of the RSD curves obtained from DEM and the Markov process, a Monte-Carlo method is

employed to move each particle from state to state by means of the transition matrix $\mathbf{P}$. More precisely, the state containing the particle p at iteration $n+1$, from state i at iteration n , corresponds to the smallest $j \in S$ such that

$$\chi(p) \leq \sum_{k=1}^j \mathbf{P}_{ik}, \quad (12)$$

where $\chi(p)$ is a random number between 0 and 1. To investigate the evolution of mixing beyond the learning time T , the map can be applied successively for a number of iterations, N_{tf} , that is for a simulation time of $T_{\text{f}} = N_{\text{tf}}\tau$.

3. Results and discussion

The proposed method was applied to the mixing of 48,000 2-mm spherical particles in a cylindrical drum having a diameter of 10 cm, a length of 9 cm and rotating at 10 RPM. These characteristics lead to a Froude number ($Fr = \omega^2 r / g$, with ω and r the angular velocity and the drum radius, respectively) of the order of 10^{-2} , which is a typical value for a rolling regime (Mellmann,

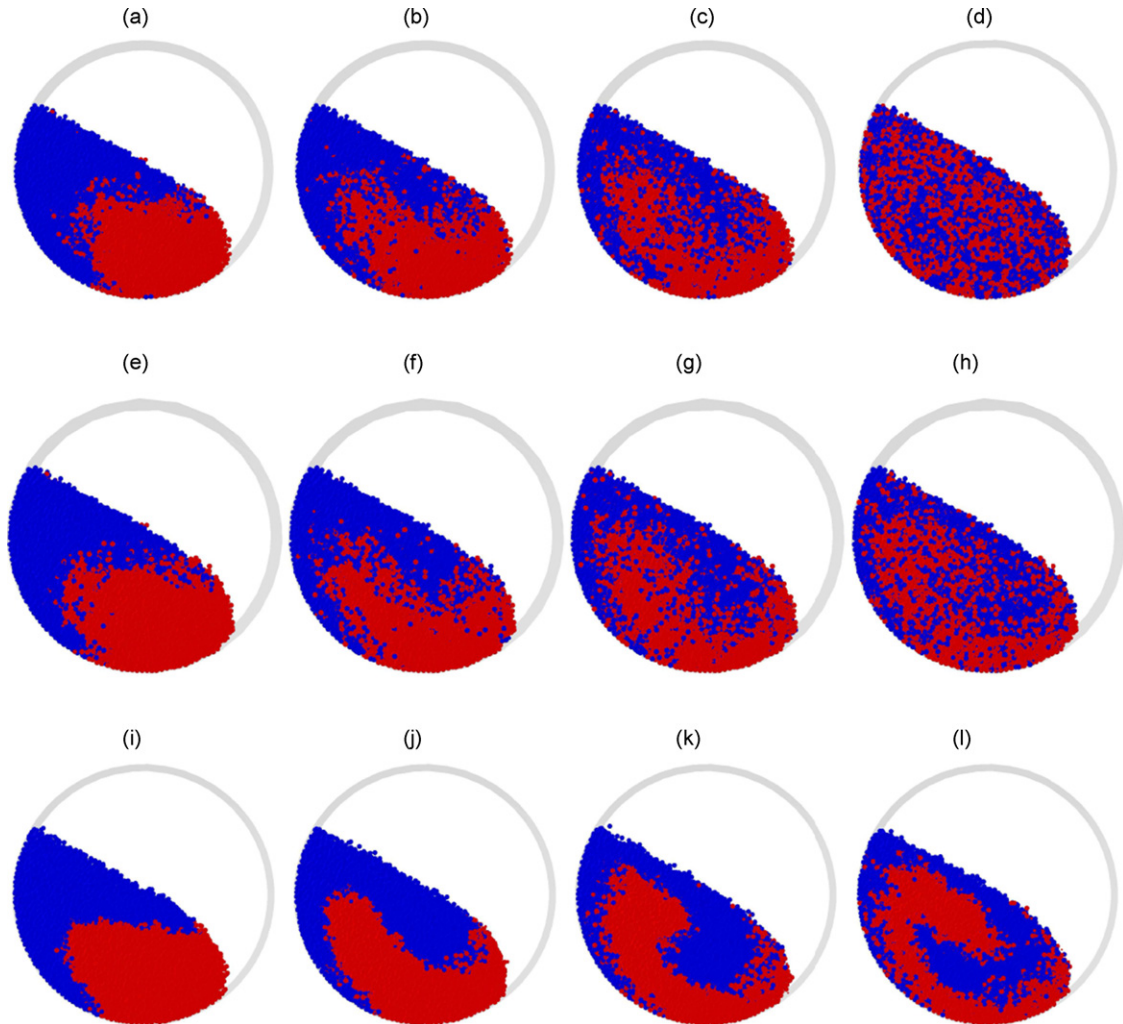


Fig. 3. Particle distributions on the middle cross-section of the cylindrical tumbler after $t = 2, 6, 10, 14$ s ($1/3, 1, 5/3, 7/3$ rev.), respectively, with $N = 1813$ and $N_{\text{LT}} = 2$. (a–d) $\tau = 2$ s; (e–h) $\tau = 4$ s; (i–l) DEM results.

2001). The computational domain was divided into N cells, each of which corresponds to one state of the system. N was varied from 72 to 5595, as discussed later. This is a classical benchmark problem that has been considered by many authors from the mixing community. Despite its apparent simplicity, this system is characterized by axial diffusion (Ding, Seville, Forster, & Parker, 2001; Sherritt, Chaouki, Mehrotra, & Behie, 2003) as well as radial mixing dynamics that leads to the formation

of a lobe within the bed of particles (Fiedor & Ottino, 2005; Jain, Ottino, & Lueptow, 2005; Khakhar, McCarthy, & Ottino, 1997; McCarthy, Khakhar, & Ottino, 2000). The velocity profile and the corresponding streamlines in permanent regime are presented in Fig. 1(b). The reader is referred to (Ding et al., 2001; Fiedor & Ottino, 2005; Mellmann, 2001; Sherritt et al., 2003) for an in-depth analysis of the particle dynamics in such a system.

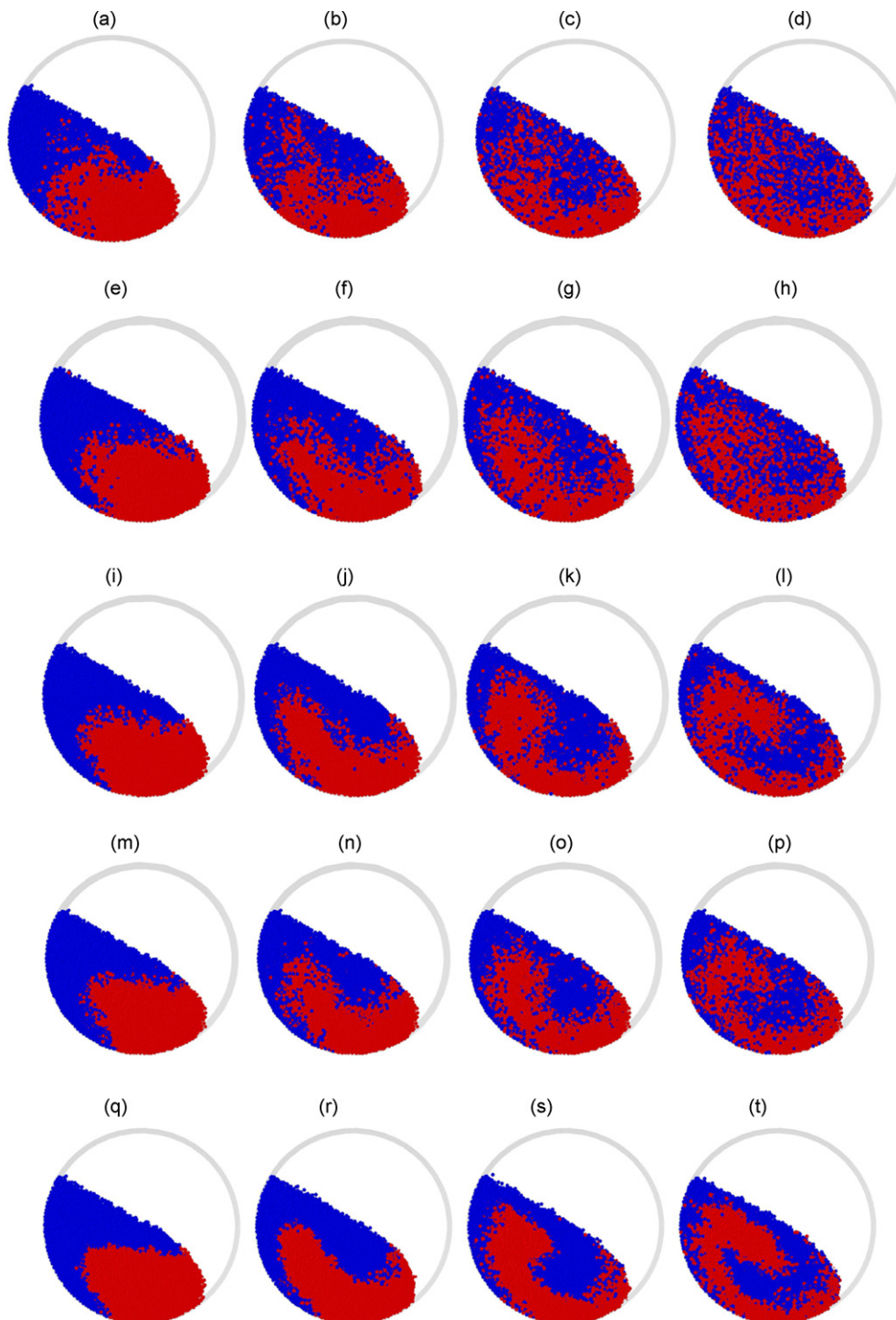


Fig. 4. Particle distributions on the middle cross-section of the cylindrical tumbler after $t = 2, 6, 10, 14$ s ($1/3, 1, 5/3, 7/3$ rev.). (a–p) Obtained using DEM–MPM ($\tau = 4$ s, $N_{LT} = 2$) with $N = 253, 1813, 3587$ and 5595 cells, respectively. (q–t) Obtained with DEM.

The DEM computations were performed in parallel using 8 Intel Xeon 3.6 GHz processors of a 64-processor Beowulf cluster, whereas the DEM-MPM computations were performed in serial mode. Relevant parameters used for this problem are gathered in Table 1. The DEM-MPM required from 1 to 10 min of CPU time for 8 revolutions of the tumbler, depending on the number of states, which is much less than the DEM simulations that required about 24 h of CPU time per processor. The granular material was originally at rest in the domain and the transition matrix was constructed when a pseudo-steady state was reached. The other DEM parameter values, the knowledge of which is not deemed essential in the context of this work, are similar to those recently used in (Lemieux et al., in press). The DEM-MPM gives the number of red and blue particles in each state at each time iteration, but does not provide the exact location of these particles. To enable the visualization of the particles in Ω , the red and blue particles were first randomly distributed within each state, which implies that the particles are assumed to be perfectly mixed within a state or a cell.

The effect of the Markov process time step, the number of states (or cells) and learning time on the accuracy of the DEM-MPM solution is next investigated. This is done by comparing the segregation patterns and RSD curves obtained with DEM-MPM to those obtained with DEM.

3.1. Effect of the Markov process time step

Fig. 2 displays RSD curves obtained using DEM and DEM-MPM with $N = 1813$, $N_{LT} = 2$ and different values of τ . As can be seen, there exists a value of τ above which the RSD curves obtained with DEM-MPM are fairly close to that obtained with DEM. In fact, this threshold corresponds to the characteristic time of the system, which can be provided by the power spectrum of the RSD. This specific time is 4.05 s, which is less than the time to achieve one complete revolution of the tumbler, that is 6.00 s at 10 rpm. This can be viewed as the period over which the system repeatedly undergoes a constant and well-defined transformation that is described in Eq. (6) as a linear map driven by the transition matrix $\mathbf{P}$.

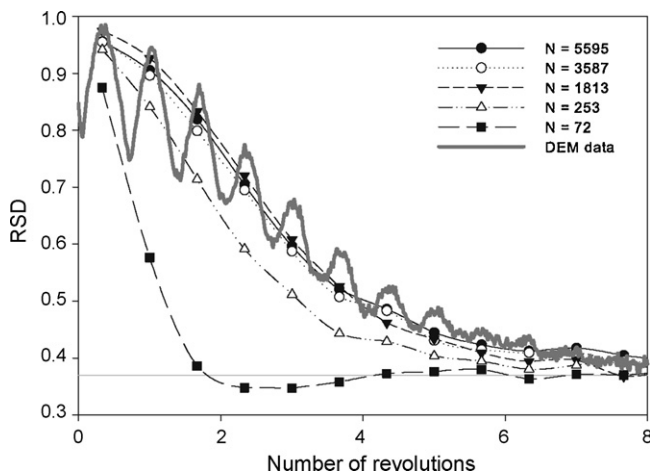


Fig. 5. Effect of the number of states N on the accuracy of DEM-MPM.

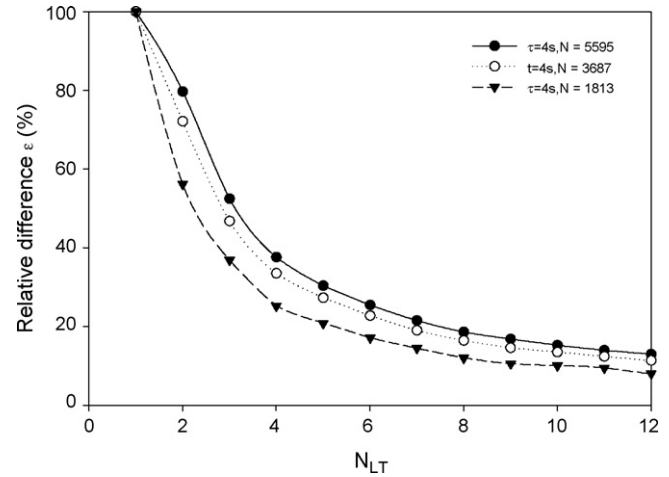


Fig. 6. Graph of ϵ versus N_{LT} for $N = 5595, 3587$ and 1813 , and $\tau = 4$ s.

Two different RSD curves obtained with $\tau = 4$ s are presented in Fig. 2(b), which result from the use of two different initial conditions. The white circle curve corresponds to the Markov process simulation using as initial condition the position of the particles at the beginning of the learning process with DEM, whereas for the triangle curve, the initial condition is delayed by two seconds. The results obtained show that, despite the fact that the Markov process cannot provide the particle positions within one time interval $]m\tau, (m+1)\tau[$ for $m \in N$ since it is discrete, its application with one single transition matrix and two different initial conditions in the same time interval $]0, \tau[$ yields results that lie within the envelope of the DEM RSD curve.

Also, one may readily see in Fig. 3 that there is good qualitative agreement between the results obtained with DEM-MPM and those obtained with DEM alone, the accuracy in the prediction of the mixing patterns being sensitive to the time step τ of the Markov chain. The apparent numerical diffusion inherent to the DEM-MPM solutions of Fig. 3 comes in part from the fact that the red and blue particles within one cell are perfectly mixed and their exact position is not known. In fact, it will be shown in the next sub-sections that the number of states N and

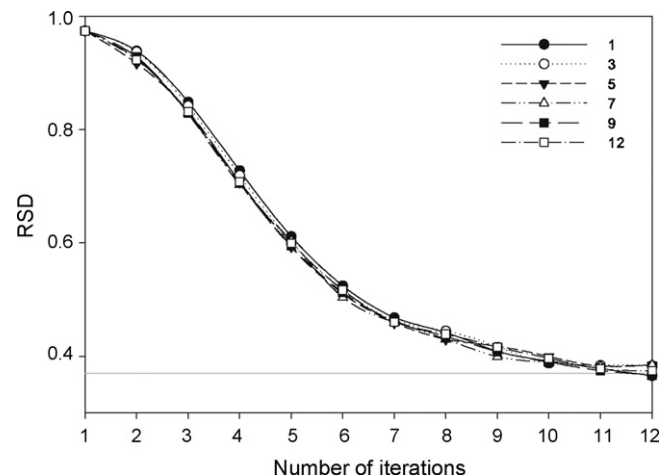


Fig. 7. Effect of N_{LT} on RSD curves with $\tau = 4$ s and $N = 1813$.

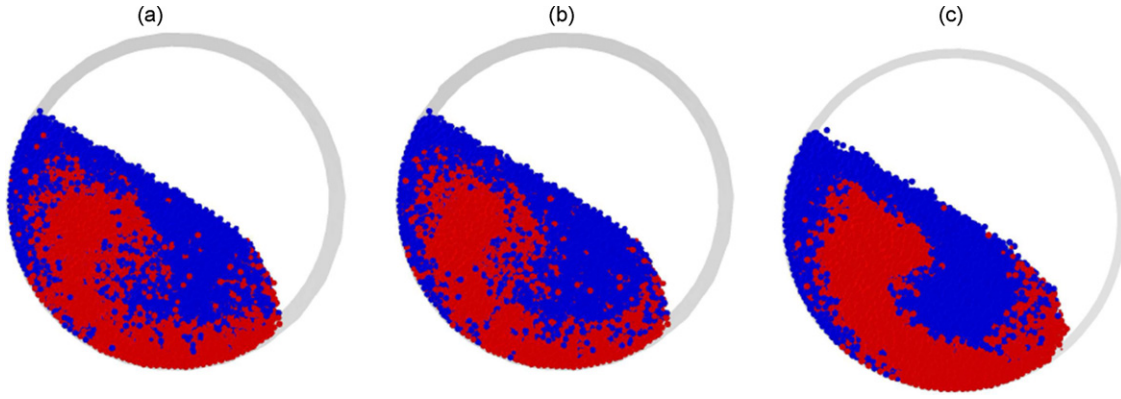


Fig. 8. Particle distributions on the middle cross-section of the cylindrical tumbler after $t = 6$ s (1 rev.) with $N = 3587$, $\tau = 4$ s. (a) $N_{LT} = 1$; (b) $N_{LT} = 12$; (c) DEM results.

the learning time T also have a strong impact on the accuracy of DEM-MPM.

3.2. Effect of the number of states

The effect of the number of cells on the level of numerical diffusion was assessed by considering 5 different decompositions of the computational domain with a decreasing cell size, as summarized in Table 1. For these simulations, the following numerical parameters were used: $\tau = 4$ s and $N_{LT} = 2$.

Fig. 4 displays the particle color distributions on the middle cross-section of the cylindrical tumbler after $t = 2, 6, 10$ and 14 s ($1/3, 1, 5/3, 7/3$ rev.). The DEM results are presented in Fig. 4(q–t). The DEM-MPM results are presented in Fig. 4(a–p) for $N = 253, 1813, 3587$ and 5595 cells, respectively. Note that the particle color distributions obtained with different cell sizes were projected onto one single reference mesh (average cell size of 1.0 cm) so that the asymptotic RSD value, RSD_r , is the same for all these curves (i.e. 0.37 ± 0.02 , as explained in Section 2.3). These figures reveal that the mixing patterns predicted with DEM-MPM converge towards those of DEM as the number of cells is increased. The same conclusion can be drawn from Fig. 5 that displays RSD curves for different values of N .

3.3. Effect of the learning time

To evaluate the effect of the learning time $T = N_{LT}\tau$ on the transition matrix $\mathbf{P}$, we denote by $\mathbf{P}(N_{LT})$ the matrix obtained by applying Eq. (8) to compute each of its entries. We then define the relative difference $\epsilon(N_{LT})$ between $\mathbf{P}(N_{LT})$ and $\mathbf{P}(N_{LT} + 1)$ by

$$\epsilon(N_{LT}) = \frac{\|\mathbf{P}(N_{LT}) - \mathbf{P}(N_{LT} + 1)\|_1}{\|\mathbf{P}(N_{LT})\|_1}. \quad (13)$$

For a matrix $\mathbf{M}$, we define the L1-norm as $\|\mathbf{M}\|_1 = \sum_{ij} |M_{ij}|$. The graph of ϵ versus N_{LT} for $\tau = 4$ s is presented in Fig. 6 for different cell sizes. It shows that the relative difference decreases exponentially for increasing values of N_{LT} , suggesting that the entries of the transition matrix converge to constant values. It is shown in Fig. 6 that the relative difference is smaller for

coarse meshes. This comes from the fact that there are more particles in larger cells, which contributes to stabilize the matrix more rapidly since more transitions per state are sampled at each iteration of the learning process. Indeed, an average of N_p/N particle transitions come into play at each iteration. The tolerance for ϵ for which it is deemed acceptable to work with the matrix should be set carefully. In fact, even if the matrix stabilizes, there is no information regarding its irreducibility and aperiodicity, which constitutes a sufficient condition for having a convergent Markov process (Chung, 1967). In addition, it must be emphasized that the stabilization of the transition matrix is not sufficient to guarantee accurate mixing predictions. This is evidenced by the results obtained on the coarse meshes in Fig. 4(d and h).

In the case of the tumbler considered in this work, Fig. 7 shows that the RSD curves are insensitive to an increase of the learning time T , and that one single iteration of the learning process suffices. Similar results (not shown here) are observed when using a larger number of cells, i.e. with $N = 3587$ and $N = 5595$. The same conclusions can be drawn by looking at the mixing patterns, such as those in Fig. 8 on the middle cross-section of a drum at $t = 6$ s, for $N_{LT} = 1$ and $N_{LT} = 12$ ($\tau = 4$ s and $N = 3587$).

4. Conclusion

In this paper, we introduced a computer model based on a stationary Markov process for the simulation of the mixing of monodisperse particles with identical properties. In DEM-MPM, the transition matrix is built using a few iterations of a DEM simulation for determining with good accuracy the displacements of the particles in a mesh of cells forming the states of the process. A central advantage of this method comes from its low CPU cost. In fact, we showed that the DEM-MPM could successfully simulate the complex mixing dynamics of monodisperse particles in a drum tumbler at a fraction of the CPU time taken up by DEM. The results obtained show the potential of this method for the development of less computationally intensive approaches in granular mixing.

The effect of the time step of the Markov process, the number of states and the learning time were investigated in the case of a cylindrical tumbler. The number of cells (states) was shown to play a significant role in the accuracy of the method. The method is consistent, the solution converges to that obtained by DEM for an increasing number of states. Finally, it was observed that the learning time of the transition matrix does not affect significantly the accuracy of the RSD if a sufficient number of transitions is sampled in each cell.

In summary, this work, which is restricted to monodisperse particles and based on stationary Markov processes, must be viewed as a first step towards the building of a hybrid DEM-based Markov process method for granular mixing. Future work will address the extension of the proposed technique to the case of polydisperse systems of particles with different properties using nonstationary Markov processes. The link between the mixing behavior of a system of particles and the properties of the corresponding Markov transition matrix are the topics of a forthcoming paper. Finally, it would be of interest to use experimental data obtained by non-intrusive methods, such as radioactive particle tracking and positron emission particle tracking, for the construction of the transition matrix (see (Doucet et al., in press; Kuo et al., 2002) for typical applications).

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